

Lifshitz–Kosevich Theory of Anomalous Landau Levels in Topological Flat Bands

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In conventional metals, quantum oscillations arise from Landau quantization of Fermi-surface cyclotron orbits, whose dynamics are governed by the Fermi velocity and cyclotron effective mass within Lifshitz–Kosevich (LK) theory. A perfectly flat band, by contrast, has vanishing group velocity, which would naïvely imply an infinite cyclotron mass and complete thermal suppression of quantum oscillations. Yet topological flat bands can support anomalous Landau levels (LLs) whose finite-field spacing is generated by quantum geometry rather than band curvature, allowing quantum oscillations to persist. This work addresses how such anomalous flat-band LLs behave within the LK framework and whether their thermal damping can reveal quantum-geometric information. Using a minimal model with exactly flat topological bands, we derive an LK theory for these anomalous LLs and analyze fixed-density magnetization oscillations. The resulting oscillations exhibit a finite LK effective mass that is substantially larger than the normal-band value and possesses a strong magnetic-field dependence. In the weak-field limit, this anomalous mass reflects the quantum-geometric origin of the LL spacing and scales inversely with both the magnetic field and the trace of the quantum metric. Thus, thermal damping of flat-band quantum oscillations directly measures the quantum metric, establishing quantum oscillations as a probe to flat-band quantum geometry.

Introduction - Quantum oscillations are conventionally understood as a Fermi-surface phenomenon [1]: in a magnetic field, cyclotron orbits are quantized into Landau levels, and their passage through the Fermi energy produces oscillatory thermodynamic and transport responses. The standard Lifshitz–Kosevich (LK) formalism [1, 2] expresses the oscillatory part of the grand potential $\delta\Omega$ at a finite temperature as a sum over harmonics,

$$\delta\Omega \propto \sum_{p=1}^{\infty} \frac{R_T(p)}{p^2} \cos(2\pi p F/B - \phi), \quad (1)$$

where p is the harmonic index, F and ϕ define the oscillation frequency and phase, respectively, and B is external magnetic field. The thermal damping factor $R_T(p) = X_p / \sinh X_p$ depends on $X_p = 2\pi^2 p k_B T / \hbar \omega_c$, with the cyclotron energy $\hbar \omega_c = e \hbar B / m^*$ set by the cyclotron mass m^* , which therefore fixes the characteristic temperature scale of thermal smearing. In systems with vanishing band dispersion, however, the band mass diverges. Applied literally to a perfectly flat band, $m^* \rightarrow \infty$ gives $X_p \rightarrow \infty$ and $R_T \rightarrow 0$, so the conventional LK formula predicts complete thermal suppression and a total absence of quantum oscillations. This conclusion, however, rests on identifying the thermal damping scale with a cyclotron energy set by the band dispersion—an identification that need not hold for topological flat bands, which have recently been realized in moiré materials [3–20]. Whether quantum oscillations persist in such a band, and how the LK damping factor should be reformulated when the conventional dispersive cyclotron energy is absent, remain unresolved.

Recently, it was recognized that the Landau levels of a

perfectly flat band under magnetic fields may not remain degenerate but instead can spread over a finite energy window. The magnitude of this anomalous Landau-level spreading is set by the quantum geometry of the Bloch states [21–25]—quantified by the Hilbert–Schmidt quantum distance and the cross-gap Berry connection—rather than by any band dispersion [26–33]. This spreading endows even a perfectly flat band with a finite, field-dependent energy scale that has no counterpart in the dispersive cyclotron picture, reopening the possibility that quantum oscillations survive in the flat-band limit—now controlled by quantum geometry rather than by band curvature. This raises two questions for topological flat bands: what replaces the cyclotron energy in the LK thermal damping of anomalous LL oscillations, and can that damping be used to extract quantum-geometric information?

To address these questions, we develop an LK theory for quantum oscillations of anomalous LLs in a minimal model with perfectly flat topological bands [34]. Our central claim is that the LK thermal damping for anomalous LL oscillations is controlled by the *local* LL spacing at the Fermi level, $v_{\mu}^{\alpha,\eta} = |\partial E_{\eta}^{\alpha}(n, B) / \partial n|_{n_{\alpha,\eta}^*}$, where E_{η}^{α} is the branch-resolved Landau-level energy and n is the Landau-level index treated as a continuous variable. We compare fixed-density magnetization oscillations of normal LLs, associated with dispersive bands, with anomalous LLs from topological flat bands. We compute their temperature dependence, extract the thermal damping from local oscillation windows, and express the results in the conventional effective-mass framework. Although the anomalous LL oscillations exhibit much stronger thermal damping, the fitted effective masses remain finite,

are approximately an order of magnitude larger than the normal-band masses in the representative comparison, and exhibit a clear monotonic field dependence, in contrast with the nearly constant normal-band masses. Expanding the exact LL spacing in the weak-field limit yields $v_\mu^{\alpha,-} \sim B^2 \text{tr } g$, so that $m_{\text{eff}}^* \sim 1/(B \text{tr } g)$, where B is the magnetic field and g is the quantum metric. The LK thermal scale of anomalous oscillations is thus directly tied to the quantum metric of the topological flat band. Taken together, these results answer both questions: (i) anomalous flat-band oscillations persist within the LK framework, with the local LL spacing $v_\mu^{\alpha,\eta}$ replacing the cyclotron energy and yielding a finite, strongly field-dependent effective mass larger than the dispersive-band value; and (ii) their thermal damping provides a probe of flat-band quantum geometry.

Ideal topological flat bands - We start with a minimal model with perfectly flat topological bands, which can describe low-energy physics of moiré heterostructures with type-II band alignment in Ref. [34]. This model consists of heavy (flat) fermion and light (dispersive) fermion components, and the corresponding Hamiltonian is given by

$$H_\alpha(\mathbf{k}) = \begin{pmatrix} E_0 + ak^2 & \alpha ck_\alpha \\ \alpha ck_{-\alpha} & E_0 + \Delta_f \end{pmatrix}, \quad (2)$$

where $\alpha = \uparrow, \downarrow$ or $\pm$ labels two opposite spin states, $k_\pm = k_x \pm ik_y$, a defines the band-curvature coefficient of the dispersive bands, c is the coupling between the flat and dispersive bands, and E_0 is the energy reference and Δ_f is the energy separation between the flat and dispersive bands at $\mathbf{k} = 0$. The above Hamiltonian is valid within the momentum cutoff Λ , e.g. $|\mathbf{k}| \leq \Lambda$, which is set by the moiré Brillouin zone size in moiré heterostructures. In our numerical results, energies and the thermal scale $k_B T$ are measured in eV and lengths in nm. Two spin Hamiltonians $H_\uparrow$ and $H_\downarrow$ are related by time-reversal (TR) symmetry, e.g. $H_\downarrow(\mathbf{k}) = H_\uparrow^*(-\mathbf{k})$, so the full system satisfies TR. When the condition $\Delta_f = \frac{c^2}{a}$ is satisfied, the lower band is perfectly flat with $E_-^{(\downarrow)}(\mathbf{k}) = E_0$ and the upper band is dispersive with a finite gap, given by $E_+^{(\downarrow)}(\mathbf{k}) = E_0 + \frac{c^2}{a} + ak^2$, as shown in Fig. 1a. The full effective Hamiltonian possesses both TR and inversion symmetries, so the eigen-states for spin up and down Hamiltonian are degenerate at each momentum. For the perfectly flat lower band $E_-^{(\downarrow)}$, its wavefunction carries nontrivial quantum geometry, as demonstrated by the Berry-curvature distribution in Fig. 1b. In particular, the lower band satisfies the ideal quantum geometry condition

$$\text{tr } g_\alpha(\mathbf{k}) = |\Omega_\alpha(\mathbf{k})| = \frac{2c^2}{(ak^2 + c^2/a)^2}. \quad (3)$$

Thus, the two TR-related sectors have the same quantum metric and opposite Berry curvature; the spin-up

(spin-down) flat band has Chern number +1 (−1). More discussion on the quantum geometry of ideal topological flat bands can be found in Sec. A of Supplementary Materials (SM).

We next consider this minimal model under a perpendicular magnetic field $\mathcal{B}$, which enters through minimal coupling $\mathbf{k} \rightarrow \boldsymbol{\Pi} = \mathbf{k} + e\mathbf{A}/\hbar$, where $\mathbf{A} = \frac{\mathcal{B}}{2}(-y, x, 0)$ is the vector potential for the symmetric gauge. The canonical momentum operators satisfy $[\Pi_x, \Pi_y] = -iB$ with $B = e\mathcal{B}/\hbar$, so the ladder operators can be defined as $b = \Pi_-/\sqrt{2B}$, $b^\dagger = \Pi_+/\sqrt{2B}$ with $[b, b^\dagger] = 1$. Here, B is measured in nm^{-2} . Using these operators, we solve the Hamiltonian in Eq. (2) (see SM Sec. B). The spectrum in Fig. 1(c) comprises two zeroth LLs,

$$E_0^\uparrow(B) = E_0 + aB, \quad E_0^\downarrow(B) = E_0 + \frac{c^2}{a} = E_0 + \Delta_f, \quad (4)$$

and four LL branches for $n \geq 1$, given by

$$E_\pm^\uparrow(n, B) = E_0 + \frac{1}{2} \left[U_n \pm \sqrt{U_n^2 - 4c^2 B} \right], \quad (5)$$

with $U_n \equiv c^2/a + (2n+1)aB$, and

$$E_\pm^\downarrow(n, B) = E_0 + \frac{1}{2} \left[V_n \pm \sqrt{V_n^2 + 4c^2 B} \right], \quad (6)$$

with $V_n \equiv c^2/a + (2n-1)aB$. We note that the zero mode $E_0^\uparrow$ linearly increases with B , while $E_0^\downarrow$ is independent of B . Moreover, the two lower-energy branches ($E_-^\uparrow$, $E_-^\downarrow$) form the anomalous LL manifold, originating from topological flat-bands as $E_-^{\uparrow,\downarrow}(B \rightarrow 0) = E_0$, while the two higher-energy branches ($E_+^\uparrow$, $E_+^\downarrow$) are the normal LLs from dispersive bands. The gaps between adjacent anomalous LLs, defined as $\Delta E_n^{\uparrow(\downarrow)} = E_-^{\uparrow(\downarrow)}(n+1) - E_-^{\uparrow(\downarrow)}(n)$, scales as $\Delta E_n^{\uparrow(\downarrow)} \propto B^2$ for a small B , slower than the linear-in- B growth of the normal LLs from dispersive bands. For a large n , we can treat n as a continuous variable to express the Fermi-level local LL spacing as

$$v_\varepsilon^{\alpha,\eta} = \left| \frac{\partial E_n^\alpha(n, B)}{\partial n} \right|_{n(\varepsilon)}, \quad E_n^\alpha(n, B) = \varepsilon, \quad (7)$$

which plays a crucial role in the thermal damping of the LK theory described below.

Quantum oscillations of anomalous LLs - We apply the standard LK theory [1, 2] to our model with both normal and anomalous LLs (see SM Sec. C) and derive the oscillatory part of the grand potential as

$$\delta\Omega = \frac{D}{2\pi^2} \sum_{\alpha,\eta} \sum_{p=1}^{\infty} \frac{v_\mu^{\alpha,\eta}}{p^2} R_T \cos[2\pi p n_{\alpha,\eta}^*], \quad (8)$$

where $D = B/2\pi$ is the LL degeneracy, the LK thermal damping factor is given by

$$R_T(p) = \frac{X_p}{\sinh X_p}, \quad X_p = \frac{2\pi^2 p k_B T}{v_\mu^{\alpha,\eta}}, \quad (9)$$

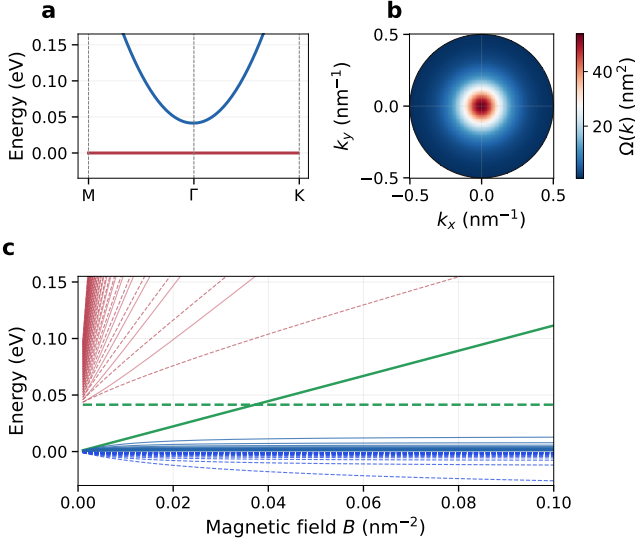


FIG. 1. (a) Band dispersion at the ideal flat-band point along the M - Γ - K path; the dispersive and flat bands are blue and red, respectively. (b) Berry-curvature distribution of the spin-up flat band. (c) Landau-level spectrum; normal, anomalous, and zeroth LLs are red, blue, and green, respectively, while solid and dashed curves distinguish the two spin sectors. Parameters: $a = 1.115 \text{ eV} \cdot \text{nm}^2$, $c = 0.215 \text{ eV} \cdot \text{nm}$, $E_0 = 0 \text{ eV}$.

and the LL filling $n_{\alpha,\eta}^*$ is fixed by the chemical potential μ through $E_{\eta}^{\alpha}(n_{\alpha,\eta}^*, B) = \mu$. Equations (8) and (9) show that the thermal damping is controlled by the local spacing $v_{\mu}^{\alpha,\eta}$ at the chemical potential. In our numerical calculations, we fix the carrier density ρ , which determines μ and $n_{\alpha,\eta}^*$ at each magnetic field $B = e\mathcal{B}/h$. The magnetization is evaluated as $M = -(e/h) \partial\Omega/\partial B$ at fixed ρ .

Figures 2(a) and 2(b) show the magnetization $M_{\mathcal{B}}$ as a function of $1/B$ for normal LLs in the electron-doped regime, $\rho = +0.01 \text{ nm}^{-2}$, and anomalous LLs in the hole-doped regime, $\rho = -0.15 \text{ nm}^{-2}$, respectively. Subtracting a smooth background isolates the oscillatory magnetization in Figs. 2(c) and 2(d). The anomalous LLs retain visible oscillations at high field, but their amplitude decays much more rapidly than that of the normal LLs as B decreases. The FFT peaks in Figs. 2(e) and 2(f) agree with the fundamental counting frequency $F_{\rho} = 2\pi|\rho_S|$, where ρ_S is the active-carrier density in the electron, shallow-hole, or deep-hole sector (see SM Sec. D). Although a Fermi-surface area is not defined for the flat band, its oscillation frequency still follows from LL counting: each filled LL accommodates $D = B/(2\pi)$ carriers per unit area, so $n^* \simeq 2\pi|\rho_S|/B$. The agreement between the FFT peaks and the dashed counting references therefore does not depend on a band dispersion. Additional frequency diagnostics are presented in SM Sec. D.

Thermal damping and effective mass - Finite tem-

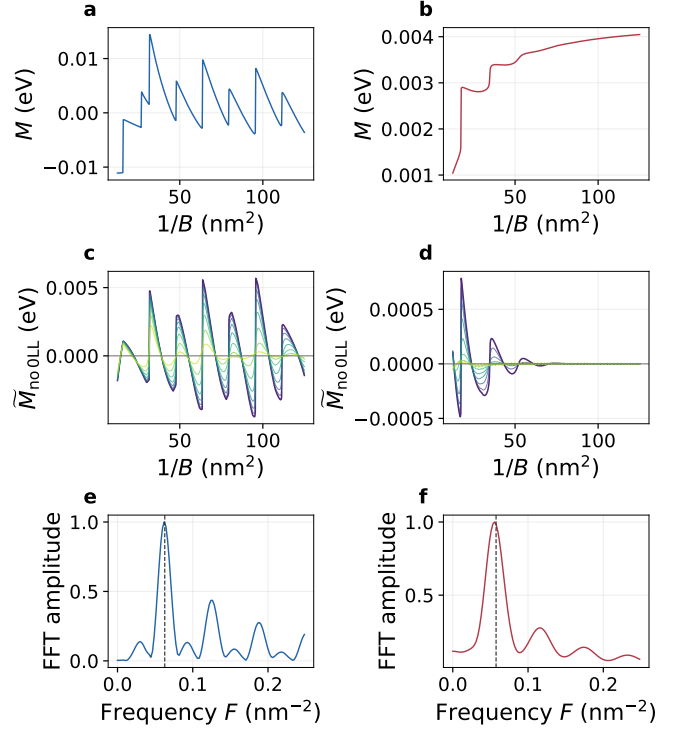


FIG. 2. Fixed-density magnetization oscillations. (a) Full physical-field magnetization $M_{\mathcal{B}}$ in the normal regime, $\rho = +0.01 \text{ nm}^{-2}$. (b) Full physical-field magnetization in the anomalous regime, $\rho = -0.15 \text{ nm}^{-2}$. (c,d) Corresponding background-subtracted oscillatory magnetization at $k_B T = (2.00, 3.07, 4.71, 7.22, 11.1, 17.0, 26.1, 40.0) \times 10^{-4} \text{ eV}$, ordered from purple to yellow. (e,f) Corresponding normalized FFT spectra; dashed vertical lines mark the active-density counting frequency $F_{\rho} = 2\pi|\rho_S|$, measured in nm^{-2} . Blue and red denote the normal and anomalous regimes, respectively.

perature suppresses the magnetization oscillations in Figs. 2(c) and 2(d). Because the signals are not perfectly sinusoidal, a global Fourier transform does not cleanly isolate the thermal damping of individual harmonics. We therefore use the short $1/B$ windows W_0 , W_1 , and W_2 marked in Figs. 3(a) and 3(b), corresponding to the first three complete high-field oscillation periods for the normal and anomalous LLs. Within each window, we extract the amplitude A_1 of the fundamental $p = 1$ harmonic and plot $A_1(T)/A_1(T_{\min})$ in Figs. 3(c) and 3(d). The anomalous oscillations show substantially stronger thermal damping. We fit this suppression with the standard LK form using a single effective mass m_{eff}^* :

$$\frac{A_1(T)}{A_1(T_{\min})} = \frac{R_T(T, m_{\text{eff}}^*, \bar{B}_w)}{R_T(T_{\min}, m_{\text{eff}}^*, \bar{B}_w)}, \quad X_1 = \frac{2\pi^2 k_B T m_{\text{eff}}^*}{\bar{B}_w}, \quad (10)$$

where $\bar{B}_w$ is the average magnetic field in the window. The dashed curves in Figs. 3(c) and 3(d) show that this one-parameter form captures the qualitative damping

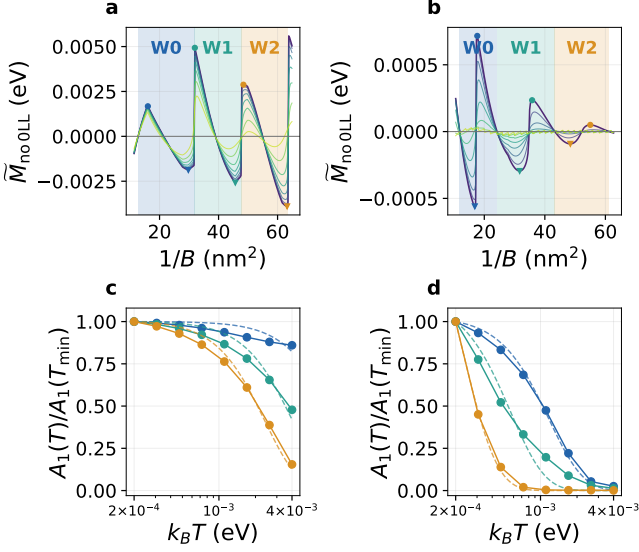


FIG. 3. Thermal damping extracted from local oscillation windows. (a,b) Oscillatory magnetization in the normal regime, $\rho = +0.01 \text{ nm}^{-2}$, and anomalous regime, $\rho = -0.15 \text{ nm}^{-2}$, respectively, at $k_B T = (2.00, 3.07, 4.71, 7.22, 11.1, 17.0, 26.1, 40.0) \times 10^{-4} \text{ eV}$, ordered from purple to yellow. In (a,b), W_0 , W_1 , and W_2 are blue, teal, and orange, respectively. (c,d) Normalized $p = 1$ harmonic amplitudes for the windows in (a,b), respectively; dashed curves are standard-LK fits.

trend in both regimes. Comparing Eq. (10) with Eq. (9) gives

$$m_{\text{eff}}^*(\bar{B}_w, \mu) = \frac{\bar{B}_w}{v_{\mu}^{\alpha, \eta}}, \quad (11)$$

in units of $\text{eV}^{-1} \text{ nm}^{-2}$. Figure 4 summarizes the extracted masses. For the normal regime ($\rho = +0.01 \text{ nm}^{-2}$), the fitted values are $m_{\text{eff}}^* \approx (0.626, 0.788, 0.958) \text{ eV}^{-1} \text{ nm}^{-2}$ across W_0, W_1, W_2 and remain comparatively small. In the anomalous regime ($\rho = -0.15 \text{ nm}^{-2}$), they increase to $m_{\text{eff}}^* \approx (5.92, 6.73, 11.0) \text{ eV}^{-1} \text{ nm}^{-2}$ and show a clear monotonic trend across the windows. The contrasting magnitude and field dependence provide an experimental signature distinguishing anomalous from normal LL oscillations.

The different trends can be understood from Fig. 1(c). At large n^* , the normal branches satisfy $(\mu - E_0)^2 \gg c^2 B$, whereas the anomalous branches satisfy $(\mu - E_0)^2 \ll c^2 B$. The normal effective mass then approaches the band-curvature value $m_{\text{eff}}^* \approx 1/(2a)$. For the anomalous branches, $m_{\text{eff}}^* \approx c^2 B/[2a(\mu - E_0)^2]$. At fixed carrier density, n^* is approximately fixed and $\mu - E_0 \propto B$ in the anomalous regime, yielding $m_{\text{eff}}^* \sim 1/B$. The dashed curves in Fig. 4 show these analytical trends and agree qualitatively with the numerical results; the derivation and additional tests are given in SM Secs. C and D.

As the spreading of anomalous LLs is directly con-

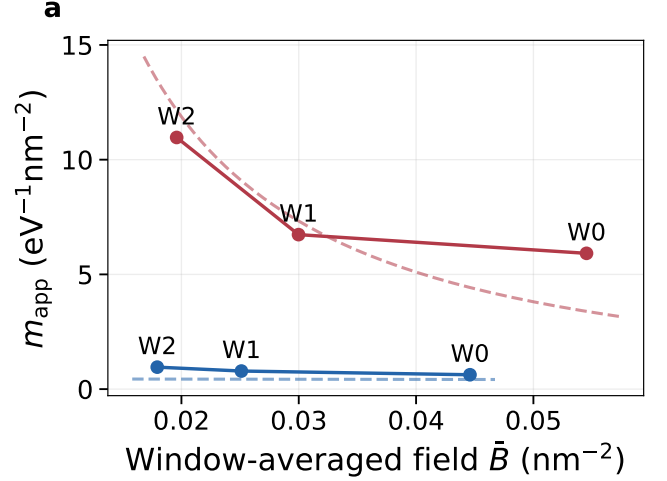


FIG. 4. (a) Apparent effective mass, measured in $\text{eV}^{-1} \text{ nm}^{-2}$, versus window-averaged magnetic field. Solid markers show masses extracted from the Fig. 3 damping fits for the normal regime, $\rho = +0.01 \text{ nm}^{-2}$ (blue), and anomalous regime, $\rho = -0.15 \text{ nm}^{-2}$ (red). Dashed curves show the corresponding analytical LK trends; labels identify the local windows.

nected to quantum geometry of topological flat bands, we finally explore how to connect the effective mass m_{eff}^* to quantum geometric information. In the semiclassical limit with a large LL filling $n_{\alpha, -}^* \gg 1$, we find $U_{n^*} \approx V_{n^*} \approx ak_{n^*}^2 + c^2/a$ with $k_{n^*}^2 \approx 2n_{\alpha, -}^* B$. Consequently, we find the anomalous-branch local LL spacing $v_{\mu}^{\alpha, -} \approx \frac{2ac^2 B^2}{(c^2/a + ak_{n^*}^2)^2} = aB^2 \text{tr } g_{\alpha}(k_{n^*})$ for both $\alpha = \uparrow, \downarrow$. This establishes the relationship between the effective cyclotron mass defined in Eq. (11) and the quantum metric,

$$m_{\text{eff}}^* \approx \frac{1}{aB \text{tr } g_{\alpha}(k_{n^*})}. \quad (12)$$

Therefore, we demonstrate the magnetic field dependence of the effective mass for anomalous LLs is directly tied to the quantum metric of topological flat bands.

Conclusion - We have developed an LK description of quantum oscillations from the anomalous Landau levels of ideal topological flat bands. The oscillations remain finite because their thermal scale is set by the local LL spacing rather than by a dispersion-derived cyclotron energy. In the representative fixed-density comparison, the anomalous oscillations damp much more rapidly than the normal oscillations and yield fitted effective masses approximately an order of magnitude larger, with a pronounced field dependence. In the semiclassical large- n^* limit, $m_{\text{eff}}^* \sim 1/[B \text{tr } g]$, directly relating the damping scale to the quantum metric. The frequency and the thermal scale thus play complementary roles: LL counting fixes the oscillation period, while quantum geometry governs the anomalous thermal damping. Given recent experimental progress in moiré materials [4, 5, 8–10, 35], temperature-dependent quantum-

oscillation measurements offer a concrete route to probing the quantum geometry of topological flat bands.

Data and code availability - The code, data, and reproduction instructions associated with this work are available in the Agentic Publication Protocol (APP) repository at <https://github.com/lcxlight/lk-anomalous-landau-levels>.

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